

What People Say And How They Move: A Causal Reanalysis of Cursor-Based Faking-Good Detection in Online Personality Questionnaires

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1 Introduction

Cursor movement is among the most ubiquitous behavioral signals available to a web interface: every modern browser exposes position, velocity, and click events through standard JavaScript handlers, and the resulting telemetry can be recorded passively without specialized hardware. HCI research has used this signal to study web-form usability (Arroyo et al., 2006; Atterer et al., 2006), to predict attention and engagement (Arapakis and Leiva, 2016; Leiva and Arapakis, 2020; Arapakis et al., 2014), and to infer cognitive and affective states the interface does not directly observe (Stillman et al., 2018; Freeman and Ambady, 2010). A high-stakes extension is detecting *deceptive self-presentation* in online personality and integrity assessments, which are deployed at scale in pre-employment screening, education, and forensic contexts where respondents face a payoff structure that rewards favorable self-presentation (Birkeland et al., 2006; Viswesvaran and Ones, 1999; Morgeson et al., 2007; Tett and Christiansen, 2007). Several recent studies report participant-level classifiers that distinguish faking-good from honest responders from cursor telemetry, with F1 in the 80–90% range (Mazza et al., 2020; Monaro et al., 2021; Kuric et al., 2025), suggesting cursor telemetry on standard web forms is approaching operational utility as a passive integrity check.

Yet this literature reports almost exclusively classification metrics—accuracy, F1, AUC—computed on datasets where honest-versus-faking labels come from an experimental manipulation. A high F1 tells us that *some* combination of cursor features separates the two groups, but not *which* features matter, how large their effects are, or whether the result would hold with a different classifier, different features, or a different sample (Yarkoni and Westfall, 2017; Rudin, 2019). Causal effect estimates on pre-specified outcomes fill

this gap: they ask how much each cursor feature actually shifts when faking is induced, attach confidence intervals, and make explicit what the study is and is not powered to detect.

This paper provides such a reanalysis of Kuric et al. (2025), a between-subjects randomized experiment in which $n = 112$ Slovak respondents completed a 48-item Big Five Inventory-2 (Soto and John, 2017; Halama et al., 2020) on their own devices using a standard one-item-per-page horizontal Likert layout, after random assignment to an Honest or Faking-Good (FG) condition; the original paper reports a participant-level XGBoost classifier with $F1 = 86.9\%$ under hierarchical aggregation of trial-level classifications. We reframe the study causally and ask: (i) *Does the FG instruction cause more socially desirable responding?*, (ii) *Do the mouse-movement metrics used to detect honesty versus fraud distinguish honest from faking-good responses?*, and (iii) *Does a history of prolonged unemployment cause more socially desirable responding?* — addressed respectively in Hypotheses 1, 2, and 3. We report inverse-probability-of-treatment-weighted (IPTW) average treatment effects with 95% confidence intervals, apply Holm adjustment to the two primary cursor outcomes and Benjamini–Hochberg FDR to the 11-feature secondary family (13 cursor features in total), and quantify the design’s minimum detectable effect (MDE). Additionally, this study serves as a pilot for a planned study of a promotion-focused, persuasive hope-appeal intervention aimed at reducing faking-good. Through this reanalysis, we ask (i) *whether less socially desirable responding is itself a marker of honesty*, and (ii) *which mouse-movement metrics, if any, could distinguish faking-good from honest responding*.

The remainder of the paper is organized as follows. Section 2 reviews how faking-good and related concepts have been studied across different paradigms and which mouse-movement metrics each literature has used. Section 3 formalizes the two primary hypotheses (the exploratory third hypothesis, H3, is developed in Appendix B). Section 4 describes the experimental setting, data, and demographic of the respondents. Section 5 reports the average treatment effects. Appendix B reports the exploratory moderation analysis. Section 6 summarizes robustness checks (detailed in the appendices). Section 7 discusses design implications of the null, and Section 8 presents gaps for follow-up work, and notes limitations.

2 Related Work

2.1 Faking-Good (FG) and Socially Desirable Responding

While it seems intuitive that FG instructions would elicit more socially desirable responses, this warrants empirical verification, considering both one’s desire to main-

tain consistency and the hypothetical nature of the scenario. According to impression-management theory, Faking-Good (FG) instructions activate respondents’ motivation to present themselves favorably by tailoring responses toward culturally valued self-presentations, exaggerating desirable traits and suppressing undesirable ones (Paulhus, 1984, 2002). Two perspectives, however, predict a weaker or more heterogeneous effect on Faking-Good instruction. Self-verification theory holds that individuals are motivated to maintain coherent, authentic self-concepts even under incentives to distort, so some respondents resist faking that threatens self-consistency (Swann, 1983). And because laboratory FG scenarios are often hypothetical, they may fail to reproduce real applicant pressure; absent meaningful consequences, strategic self-presentation may remain limited (Marcus, 2009). The net effect of FG instruction is thus an empirical question.

To counter this attenuation risk, the original study Kuric et al. (2025) adopted a “dream-job” application scenario for the FG group, mirroring the high-stakes settings where faking is common: 30–63% of respondents admit lying in real job applications to portray themselves positively (Baer and Miller, 2002; Dwight and Donovan, 2003). We test whether this simulated employment setting and the FG instruction yield more socially desirable responses. Table 1 summarizes these competing positions and the evidence motivating our design.

2.2 Misreporting: Fraud, Impression Management (Faking-Good), and Self Deception

Misreporting is a broad umbrella term that subsumes several distinct phenomena; we organize them along two axes—*material victim/gain* and *intent*—to locate Faking-Good relative to its neighboring concepts. The first axis separates *fraud* from all other forms of misreporting. Fraud involves intentional misrepresentation that produces a deceived victim and material gain at that victim’s expense (Albrecht et al., 2019); the remaining categories entail no such victim or gain. The second axis, intent, differentiates among those non-fraudulent categories. Within the Total Survey Error framework, misreporting is a component of measurement error spanning information respondents provide deliberately or unintentionally (Groves, 1989; Biemer, 2010), and the cognitive model of survey response distinguishes deliberate misreporting—distortion at the response-editing stage—from non-deliberate misreporting arising at the comprehension, retrieval, or judgment stages (Tourangeau et al., 2000). The psychometric tradition offers a parallel distinction: socially desirable responding is partitioned into *impression management*, a conscious effort to present an overly favorable self-portrait, and *self-deceptive enhancement*, an unconscious positive self-bias (Paulhus, 1984, 2002). Only impression management is intentional. Faking-Good, as operationalized in the present study, aligns with impression

Table 1: Theoretical and empirical positions on whether faking-good (FG) instructions produce socially desirable responding.

Source	Predicted effect of FG	Rationale
Impression-management theory Paulhus (1984, 2002)	Increases	FG activates motivation to tailor responses toward culturally valued self-presentations
Self-verification theory Swann (1983)	Attenuates	Respondents resist faking that threatens a coherent, authentic self-concept, even under situational incentives to distort.
Perceived-stakes account Marcus (2009)	Attenuates	Hypothetical lab scenarios fail to reproduce real applicant pressure; absent meaningful consequences, strategic self-presentation is limited.
Applicant-faking prevalence Baer and Miller (2002); Dwight and Donovan (2003)	Increases (empirical)	30–63% of respondents admit lying in real high-stakes job-application settings to portray themselves positively.
Present study Kuric et al. (2025)	Tested	Adopts a “dream-job” framing to raise stakes and counter the attenuation risk; empirically tests whether FG yields more socially desirable responses.

management: it is deliberate yet produces no material victim. Self-deception, by contrast, is non-deliberate and likewise produces no material victim. Organizing these three categories by *material victim/gain* and *intent* yields Table 2.

Table 2: A taxonomy of misreporting by intent, material victim/gain

Category	Intentional	Material victim/gain
Fraud	Yes	Yes
Faking-Good (Impression management)	Yes	No
Self-deception	No	No

Notes. “Intentional” denotes a deliberate effort to misrepresent.

Faking-good—in the sense of impression management—thus occupies a specific cell: intentional, unlike Self-Deception, but lacking both a materially harmed victim and material gain like Fraud. This positioning motivates our focus: fraud detection has been extensively studied in Information Systems (Weinmann et al., 2022) and Computer Science (Khan et al., 2024), yet how people behave when faking good—rather than committing fraud—remains largely unexplored.

2.3 Mouse Trajectory in Fraud Detection and Faking-Good

A growing body of work proposes that cursor telemetry can infer *deceptive intent*. The premise is cognitive: when more than one response is activated, movement trajectories deviate away from (Howard and Tipper, 1997) or toward (Welsh et al., 1999) a competing nontarget stimulus, reflecting parallel activation of multiple response candidates during selective reaching (Welsh and Elliott, 2004). Resolving this competition recruits executive processes, leaving traces in continuous motor output—most notably the speed and spatial deviation of the cursor path (Weinmann et al., 2022)—which has motivated a range of mouse-movement features as behavioral biometrics for fraud detection (Khan et al., 2024). The clearest fraud demonstration is Weinmann et al. (2022): relative to honest users, respondents committing fraud for financial gain moved more slowly and with greater path deviation, and the effects scaled with the extent of the fraud. The principal metrics in this literature therefore concern movement speed (e.g., total time, average velocity) and spatial deviation (e.g., average deviation from ideal path).

Studies of faking-good have predominantly been conducted under controlled laboratory conditions. Mazza et al. (2020) had 120 participants complete MMPI-2 (L, K, S) and PPI-R Virtuous Responding validity scales via mouse in a 2 x 2 design crossing instruction (honest vs. faking-good) with time pressure in a laboratory setting, finding fakers generally slower along the horizontal axis with larger maximum-deviation times, though spatial effects were inconsistent across scales. Monaro et al. (2021) extended this paradigm to a four-alternative format using PPI-R VR and PAI Positive Impression Management scales, combining temporal mouse indicators with ML classifiers, though the study likewise remained confined to laboratory conditions.

Kuric et al. (2025) move toward ecological validity by studying faking-good “in the wild”: in a between-subjects experiment where participants used their own devices remotely without laboratory standardization, respondents completed the BFI-2 through an ordinary web form while their mouse dynamics were recorded. As in the earlier studies, ground truth was set by instruction—one group was told to present themselves more favorably than they are, the other to respond honestly—but rather than relying on specialized validity items, the authors build a user behavior model from mouse-dynamics features and train classifiers to predict faking-good intent. However, across these studies the emphasis has been on detection—whether faking-good can be identified or distinguished from honest responding—rather than on establishing whether faking-good *causes* systematic changes in mouse dynamics. A further distinction concerns the naturalness of the task itself: although both Weinmann et al. (2022) and Kuric et al. (2025) were conducted online and “in the wild,” the two settings differ in kind. Kuric et al. (2025) asks respondents to complete an ordinary questionnaire—a task they would plausibly perform in everyday

life—whereas Weinmann et al. (2022) places respondents in a purpose-built decision task where they may fake to obtain financial gain. Whether and how the faking-good condition causally alters movement behavior under such naturalistic conditions thus remains an open question. Table 3 compares the context, setting, instrument, and mouse-trajectory metrics used across the studies discussed.

Table 3: Mouse-trajectory metrics across fraud and faking-good detection settings

Study	Context	Setting	Instrument	Representative trajectory metrics
Weinmann et al. (2022)	Fraud	Controlled online	Financial incentive (fraud vs. honest)	Movement speed/velocity; spatial deviation from direct path
Mazza et al. (2020)	Faking-good	Laboratory	Faking-good vs. honest; with vs. without time constraint	Response time; x-/y-axis velocity; maximum deviation; maximum-deviation time
Monaro et al. (2021)	Faking-good	Laboratory	Faking-good vs. honest	Temporal mouse indicators (e.g., initiation time, total time) feeding ML classifiers
Kuric et al. (2025)	Faking-good	Online form (in the wild)	Faking-good vs. honest	Data-driven user-model features over mouse dynamics (movement, timing, trajectory) selected for ML prediction

3 Theory and Hypotheses

We translate three questions from the Related Work into pre-specified, causally interpretable hypotheses: whether the Faking-Good (FG) instruction (H1) raises socially desirable responding, (H2) shifts the cursor metrics used to distinguish honest from dishonest responding, and (H3) interacts with prior employment adversity to deepen engagement with the faking task.

3.1 H1: FG Instruction Causally Elevates Response Social Desirability

Theoretically primary indicators. Impression-management theory holds that FG instructions activate the motivation to exaggerate desirable traits and suppress undesirable ones Paulhus (1984, 2002), and the dream-job framing of Kuric et al. (2025) heightens this by mirroring applicant settings where faking is common Baer and Miller (2002); Dwight and Donovan (2003). However, Self-verification pressures of maintaining

a constant self Swann (1983) and the attenuated stakes of hypothetical scenarios Marcus (2009) predict a weaker effect, leaving the magnitude an empirical question; we nonetheless expect the realistic dream-job framing to sustain impression-management motives and override the tendency toward self-consistency. Following Paulhus (1984), social desirability SD_i is the participant-level mean of the four BFI-2 trait scores after orientation alignment (Negative Emotionality reverse-scored), with higher values indexing more favorable self-presentation.

H1 estimand and hypothesis. The average treatment effect on social desirability is

$$ATE_{SD} = E[SD_i \mid D_i = \text{FG}] - E[SD_i \mid D_i = \text{H}], \quad (1)$$

where i indexes participants and $D_i \in \{\text{FG}, \text{H}\}$ is the randomly assigned condition; random assignment identifies ATE_{SD} as the difference between these two conditional expectations. We predict

$$\text{H1: } ATE_{SD} > 0. \quad (2)$$

H1 also serves as a manipulation check: a significant effect on questionnaire content is a necessary precondition for interpreting the H2 cursor estimates as effects of induced faking rather than of the instruction itself.

3.2 H2: FG Instruction Causally Affects Participant-Level Cursor Features

Theoretically primary indicators. If FG increases socially desirable responding, the tension between truthful and fabricated answers should induce greater cognitive conflict in the FG group, which cognitive-conflict models expect to surface in mouse telemetry: competing response candidates activate in parallel and leave traces in continuous motor output Howard and Tipper (1997); Welsh et al. (1999); Welsh and Elliott (2004); DePaulo et al. (2003); Suchotzki et al. (2017); Botvinick et al. (2001). The Response Activation Model (Welsh and Elliott, 2004) holds that intentional misreporting induces competition between response alternatives, deviating the cursor from the most efficient path, and Weinmann et al. (2022) document exactly such differences between fraudulent and honest responding. Because FG shares this core intention–response mismatch while lacking the material stakes of fraud (Section 2.2), we hypothesize that FG framing still yields distinguishable trajectories. Crucially, movement speed and spatial deviation are not fraud-specific: they are the canonical kinematic and spatial expressions of response competition shared across misreporting paradigms (cf. Mazza et al. (2020)), and we ask whether they transfer down the stakes gradient from fraud to the reputational-only setting

of FG.

We operationalize these two constructs using the corresponding item-level features released with the Kuric et al. (2025) dataset, computed for each BFI-2 item (pageview) of a participant and then averaged across that participant’s m items. For item $j \in \{1, \dots, m\}$ with captured cursor coordinates $(x_{jk}, y_{jk})_{k=1}^{n_j}$, the realized path length is

$$L_j = \sum_{k=1}^{n_j-1} \sqrt{(x_{jk} - x_{j,k+1})^2 + (y_{jk} - y_{j,k+1})^2}. \quad (3)$$

Average trajectory deviation — the mean item-level difference between the realized path and the ideal (shortest) path, where the ideal path connects the initial cursor position $\mathbf{o}_j$, the answer click $\mathbf{a}_j$, and the next-button click $\mathbf{b}_j$ (Kuric et al., 2025, Eqs. 9–10):

$$L_j^{\text{ideal}} = \|\mathbf{a}_j - \mathbf{o}_j\| + \|\mathbf{b}_j - \mathbf{a}_j\|, \quad (4)$$

$$\text{Deviation} = \frac{1}{m} \sum_{j=1}^m (L_j - L_j^{\text{ideal}}). \quad (5)$$

Average movement speed — the mean item-level realized path length divided by the movement time span Δt_j (the elapsed time from the first to the last recorded movement on item j) (Kuric et al., 2025, Eq. 18):

$$v_{\text{avg}} = \frac{\sum_{j=1}^m L_j}{\sum_{j=1}^m \Delta t_j} \quad (6)$$

These two features correspond, respectively, to the dataset columns `big5_mean_real_ideal_trajectory_diff` and `big5_mean_velocity`. `velocity` matches Weinmann et al. (2022)’s movement-speed construct directly (path length over elapsed movement time). The deviation feature captures the same path-deviation construct as Weinmann’s measure but operationalizes it as an *absolute* length difference (realized minus ideal, in pixels) rather than a normalized actual-to-shortest ratio; it therefore remains on a device-dependent pixel scale, a property we return to when interpreting its wide interval (Section 5.3).

H2 estimand and hypotheses. For each cursor outcome Y_i —instantiated as participant i ’s average trajectory deviation (Eq. 5) or average movement speed (Eq. ??)—the average treatment effect is

$$\text{ATE}_Y = E[Y_i \mid D_i = \text{FG}] - E[Y_i \mid D_i = \text{H}], \quad (7)$$

identified by random assignment and estimated separately for each outcome. Following Weinmann et al. (2022) we predict greater deviation and slower speed under FG:

$$\text{H2a: } \text{ATE}_{\text{Deviation}} = E[\text{Deviation}_i \mid D_i = \text{FG}] - E[\text{Deviation}_i \mid D_i = \text{H}] > 0, \quad (8)$$

$$\text{H2b: } \text{ATE}_{\text{Speed}} = E[\text{Speed}_i \mid D_i = \text{FG}] - E[\text{Speed}_i \mid D_i = \text{H}] < 0. \quad (9)$$

H2a follows the response activation model (Welsh and Elliott, 2004), whereby response competition pulls the cursor off the direct path; H2b follows cognitive load theory (Sweller, 1988; Buller and Burgoon, 1996), whereby the working-memory cost of fabricating a credible response slows the hand (Weinmann et al., 2022).

3.3 H3 (Exploratory): Prior Employment Adversity as a Moderator

We also pre-specify a third, *exploratory* hypothesis: that the FG effect is moderated by prior employment adversity, such that participants who have faced long-term unemployment or involuntary job loss show a larger shift toward socially desirable responding and a stronger cursor signature of faking. Because the resulting moderator subgroups are small and the interaction test is substantially underpowered at $n = 112$ (Gelman, 2015; Simmons et al., 2011), we develop H3 in full—its theory, estimand, and results—in Appendix B rather than in the main text.

4 Data and Methodology

4.1 Data Source and Authors’ Contribution

The data we analyze were collected by Kuric et al. (2025) to train and evaluate machine-learning classifiers of faking-good intent on Likert-scale personality questionnaires. Data is publicly available at <https://github.com/delieresearch/honest-faking-good-intents-forms-prediction>.

The participants, data-collection procedure, interface, and outcome measures are those described in Kuric et al. (2025). We summarize the procedure only briefly and devote more attention to how we adapt and reorganize the outcome measures from Kuric et al. (2025) for our own analyses.

4.2 Participants

Purpose-guessing exclusions. The source paper reports excluding 54 of 166 initial respondents for (a) careless responding identified by an attention-check item asking the respondent to select the option “Disagree” (Slovak: *Nesúhlasím*) as a counterintuitive option Meade and Craig (2012), (b) incomplete data, or (c) correct guessing of the study’s purpose. The dataset we analyze is the post-exclusion sample of $n = 112$; we inherit these exclusion criteria. We confirm that none of the 112 retained participants failed the attention check.

Valid participants size. The final sample comprises $n = 112$ participants (57 Honest, 55 Faking-Good), recruited by the source authors in two stages: convenience sampling via Slovak-language social media channels, followed by targeted recruitment via an online panel to improve representativeness. Mean age was 23.99 years ($SD = 7.84$); 61 male, 50 female, 1 unspecified. All participants used a computer with a mouse and consented to data collection in compliance with the General Data Protection Regulation.

4.3 Interface Design and Procedure

Interface design The study is naturalistic in two ways: participants used their *own device, browser, and physical setting*, and the interface used a *standard horizontal Likert layout with a neutral midpoint, one item per page* — a layout that closely matches what users encounter in production online assessment systems and survey platforms. This contrasts with the engineered laboratory paradigms used in much of the prior cursor-telemetry deception-detection literature (e.g., the four-quadrant full-screen layout of Monaro et al. (2021)), in which response options are spatially separated to maximize trajectory deviation. The standard layout is the more demanding test of whether cursor-telemetry inference will hold up in deployment.

Procedure Respondents began by confirming, through an on-screen prompt (Figure 1), that they were completing the survey on a device with a computer mouse rather than a touchpad or touchscreen. They then received condition-specific instructions corresponding to their assigned arm—Honest or Faking-Good—and proceeded to the personality questionnaire. The questionnaire was a 48-item subset of the validated Slovak adaptation of the BFI-2 Halama et al. (2020); Soto and John (2017), covering four of the original five Big Five domains (Extraversion, Agreeableness, Conscientiousness, and Negative Emotionality); the Open-Mindedness scale had been dropped by the source authors following pilot testing. Each item appeared on a separate page as a horizontal five-point

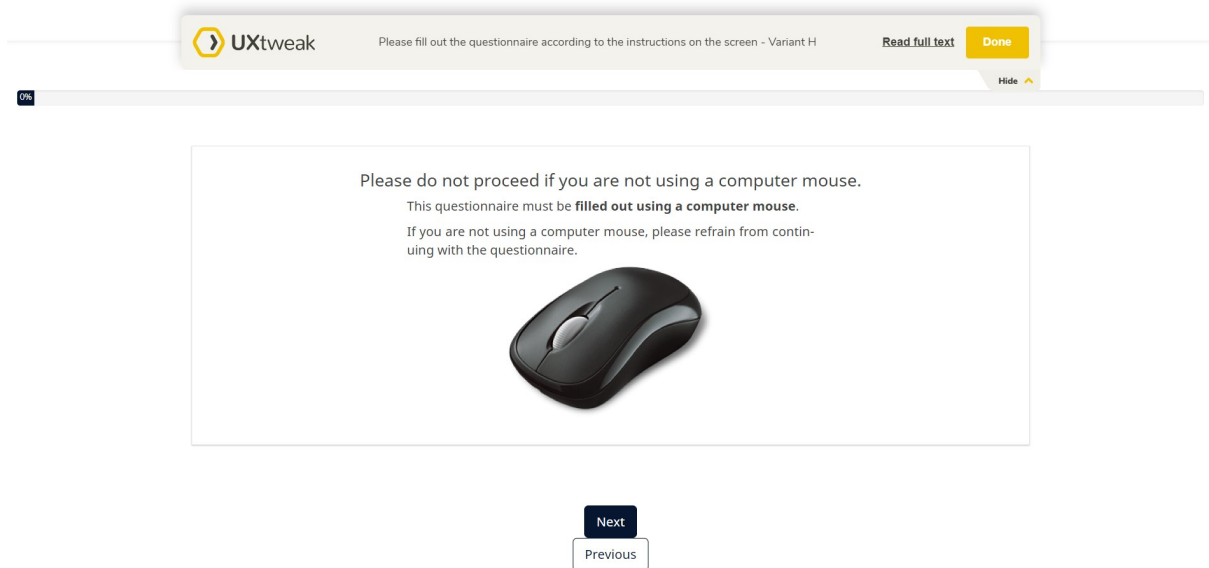


Figure 1: Mouse Alert

Likert scale, with a soft 20-second timer that, upon expiry, displayed a message and reset without further consequence.

After completing the BFI-2, respondents answered a 14-item validation and demographic block covering an attention check, self-rated computer skill, language proficiencies, marital status, age, education, employment status, household size, prior job changes, history of dismissal, history of long-term unemployment, and sex. This yielded 62 questionnaire pages per participant. Mouse interactions were captured throughout the session via the UXTweak session-recording instrument.

4.4 Random Assignment

Setting Participants were randomly assigned to one of two between-subjects conditions. In the Honest (H) condition ($n = 57$), participants were instructed to “answer truthfully and honestly” and were informed that their responses would remain anonymous. In the Faking-Good (FG) condition ($n = 55$), participants were instructed to “answer in a way that makes you seem like the best possible candidate to a future employer” and were explicitly told that, if necessary, they could lie. Examples of the honest and faking-good conditions, adapted from the original paper, are shown in Figure 2 panels (a) and (b), respectively.

4.5 Covariate Balance

The six covariates in Table 4 were pre-specified on prognostic grounds—retaining variables plausibly related to mouse-trajectory behavior (age, computer skill, sex) or to engage-

UXtweak Please fill out the questionnaire according to the instructions on the screen - Variant H Read full text Done

3%

I am someone who is outgoing, sociable

- Answer truthfully and honestly. Your answers are anonymous.
- Choose one of the following answers

Time remaining 00:00:09

Disagree strongly Disagree a little Neutral Agree a little Agree strongly

Next Previous

(a) Honest condition

UXtweak Please fill out the questionnaire according to the instructions on the screen - Variant FG Read full text Done

3%

I am someone who is outgoing, sociable

- Answer in a way that makes you seem like the best possible candidate to a future employer. If necessary, lie.
- Choose one of the following answers

Time remaining 00:00:18

Disagree strongly Disagree a little Neutral Agree a little Agree strongly

Next Previous

(b) Faking-good condition

Figure 2: Example questions under different experimental conditions

ment with the faking-good task (employment, job changes, education), and excluding demographics with no credible pathway to either (e.g., marital status, household size, language proficiency). Unlike predictive feature selection (Kuric et al., 2025), the aim here is confounding adjustment, for which non-prognostic variables add weight instability at $n=112$ without reducing bias.

Table 4 reports pre-treatment covariate balance between the Honest (H) and Faking-Good (FG) conditions using our specified variables. Two variables exhibit notable imbalance according to standard mean difference (SMD) and its corresponding p value: the proportion of female participants (H: 58%, FG: 31%; $p = .004$, $\text{SMD} = -0.56$) and self-reported computer skill (H: 3.23, FG: 3.71; $p = .033$, $\text{SMD} = +0.41$). Age shows borderline imbalance ($p = .075$, $\text{SMD} = +0.34$). Given $n = 112$, sampling-noise imbalance on at least one of six variables is not unexpected, and we address it analytically through covariate-adjusted estimates and inverse-probability-of-treatment-weighted (IPTW) estimators (Section 4.6).

Table 4: Pre-treatment covariate balance.

Covariate	Honest ($n=57$)	Faking-Good ($n=55$)	SMD	p
Age (years)	22.68 (6.01)	25.35 (9.24)	+0.34	.075
Female (proportion)	0.58 (0.50)	0.31 (0.47)	-0.56	.004
Computer skill (1-5)	3.23 (1.20)	3.71 (1.17)	+0.41	.033
Cumulative job changes (0-4)	0.44 (1.00)	0.45 (0.90)	+0.02	.929
Education (1-4)	1.91 (1.06)	1.96 (1.17)	+0.05	.808
Employment status (0-3)	1.09 (0.51)	1.29 (0.69)	+0.34	.079

Note. Cell values are mean (SD) unless otherwise indicated. $\text{SMD} = (\text{FG} - \text{H}) / \text{pooled SD}$; positive values indicate FG higher. p -values are from Welch’s t -test.

4.6 Inverse-Probability-of-Treatment-Weighted Estimators

To address the imbalances identified in Section 4.5, we IPTW weights from a logistic propensity score model that regresses the Faking-Good indicator on all six pre-treatment covariates. The target estimand is the average treatment effect (ATE); weights are stabilized to reduce variance without altering the target.

For each participant i ($n = 112$), let $T_i \in \{0, 1\}$ indicate Faking-Good ($T_i = 1$) versus Honest ($T_i = 0$) assignment, and let $X_i = (\text{age}_i, \text{female}_i, \text{comp_skill}_i, \text{jobchg}_i, \text{edu}_i, \text{emp}_i)^\top$ be the six pre-treatment covariates. We model $e(X_i) \equiv \Pr(T_i = 1 \mid X_i)$ by logistic regression,

$$\log\left(\frac{e(X_i)}{1 - e(X_i)}\right) = \beta_0 + \beta^\top X_i, \quad (10)$$

yielding fitted propensity scores $\hat{e}(X_i) = [1 + \exp(-\hat{\beta}_0 - \hat{\beta}^\top X_i)]^{-1}$. The stabilized IPTW

weights targeting the ATE are

$$w_i = T_i \frac{\hat{\pi}}{\hat{e}(X_i)} + (1 - T_i) \frac{1 - \hat{\pi}}{1 - \hat{e}(X_i)}, \quad (11)$$

where $\hat{\pi} = n^{-1} \sum_{i=1}^n T_i$ is the marginal treatment proportion. Stabilization preserves the target estimand while shrinking the variance of the weights, improving the stability of downstream estimators in small samples (Robins et al., 2000). The IPTW estimator of the ATE is

$$\hat{\tau}_{\text{IPTW}} = \frac{\sum_i w_i T_i Y_i}{\sum_i w_i T_i} - \frac{\sum_i w_i (1 - T_i) Y_i}{\sum_i w_i (1 - T_i)}, \quad (12)$$

with standard errors from a heteroskedasticity-robust (Huber–White) sandwich estimator that accounts for the estimated weights.

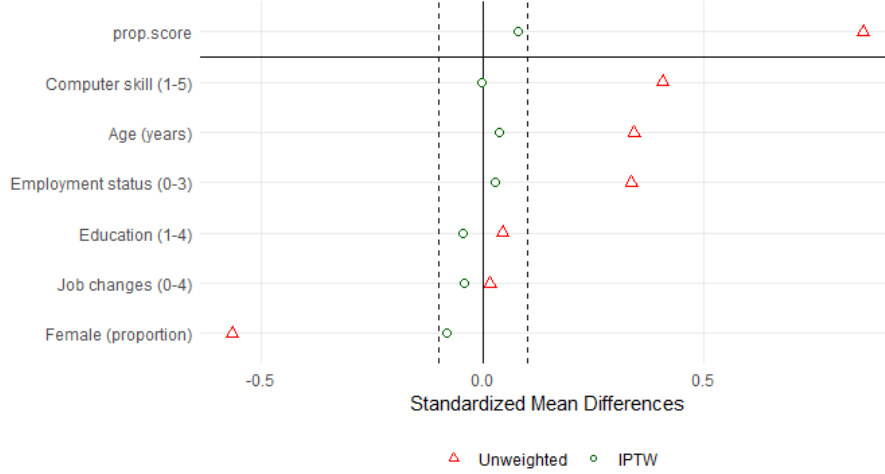
Balance after weighting. We complement the SMD with Kolmogorov–Smirnov (KS) statistics to confirm distributional, rather than merely mean-level, balance after weighting. Table 5 reports covariate balance before and after weighting. Unweighted, three covariates exceed the $|\text{SMD}| < 0.1$ threshold—female (-0.564), computer skill ($+0.407$), and age ($+0.342$)—and the propensity score shows severe separation ($\text{SMD} = +0.860$). After weighting, every covariate falls below threshold (all $|\text{SMD}| \leq 0.080$) and the propensity-score SMD shrinks by an order of magnitude to $+0.080$; the KS statistics drop in parallel, with only the propensity score remaining slightly above threshold ($\text{KS} = 0.151$).

Table 5: Covariate balance before and after IPTW.

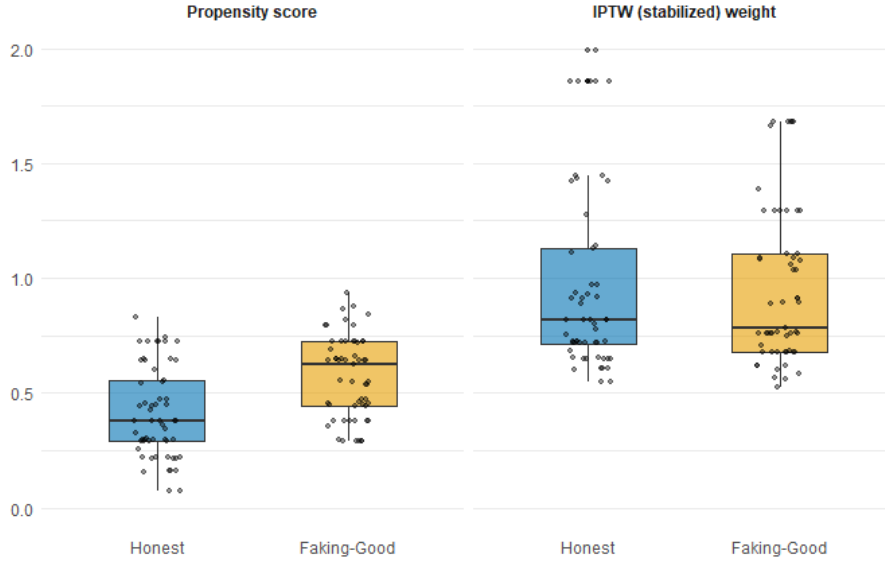
Covariate	SMD		KS	
	Unweighted	IPTW	Unweighted	IPTW
Age (years)	+0.34	+0.04	0.07	0.02
Female (proportion)	−0.56	−0.08	0.27	0.04
Computer skill (1–5)	+0.41	0.00	0.22	0.03
Cumulative job changes (0–4)	+0.02	−0.04	0.06	0.04
Education (1–4)	+0.05	−0.04	0.06	0.06
Employment status (0–3)	+0.34	+0.03	0.09	0.03
Propensity score	+0.86	+0.08	0.36	0.15

Note. SMD = standardized mean difference; KS = Kolmogorov–Smirnov statistic. The conventional threshold for adequate balance is $|\text{SMD}| < 0.1$ and $\text{KS} < 0.1$.

The Love plot (Figure 3a) shows the unweighted SMDs (triangles), several outside the ± 0.1 band, collapsing tightly around zero after IPTW (circles). Because the SMD captures only mean differences, we also compare the full propensity-score densities of the two arms using box plot and propensity score overlap plot. The box plots (Figure 3b) make the mechanism explicit: the raw propensity-score distributions differ between arms



(a) Love plot of standardized mean differences (SMDs) before and after IPTW.



(b) Distributions of the propensity score and the stabilized IPTW weights by condition.

Figure 3: Covariate balance before and after IPTW.

(Faking-Good shifted higher), whereas after weighting the stabilized-weight distributions are comparable across conditions, rendering the arms exchangeable for estimating the ATE. We could observe the same effect via the Propensity score overlap (Figure 4). Under Unweighted condition, the Honest arm concentrates at lower scores and the Faking-Good arm at higher, with overlap confined to the mid-range; after IPTW the two densities become nearly indistinguishable. The unweighted overlap—shared support across the propensity-score range—provides empirical support for positivity, while the post-weighting collapse of both the SMDs and the score densities indicates that exchangeability on measured covariates is achieved in the weighted sample. Subject to consistency and no unmeasured confounding, this justifies interpreting the IPTW-weighted contrast as an estimate of the ATE.

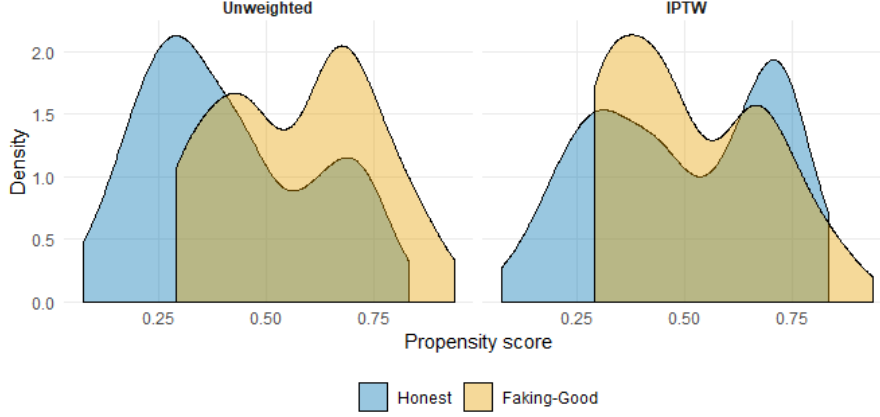


Figure 4: Propensity score overlap before and after IPTW

4.7 Measures

Self-report outcome (H1). Each BFI-2 trait domain is computed as the mean of its items per the BFI-2 scoring key, with false-keyed items reverse-coded. Negative Emotionality is reverse-scored so that all four orientation-aligned trait scores increase with social desirability (as in Kuric et al., 2025, Table 2). The participant-level outcome SD_i in Equation 1 is the mean of the four orientation-aligned trait scores; this construct corresponds to the impression-management component of Paulhus (1984)’s two-factor model of socially desirable responding.

Cursor-feature outcomes (H2). The two theoretically primary cursor outcomes are the participant-level features `big5_mean_real_ideal_trajectory_diff` and `big5_mean_velocity` from the Kuric et al. (2025) dataset, which operationalize the path-deviation and movement-speed constructs of Section 3.2. For each questionnaire item (pageview), *real-ideal trajectory difference* is the realized cursor path length minus the ideal (shortest) path connecting the initial cursor position, the selected answer, and the next-button click (Kuric et al., 2025, Eqs. 9–10), and *velocity* is the realized path length divided by the movement time span (Kuric et al., 2025, Eq. 18); each is averaged across a participant’s BFI-2 items to form the participant-level `big5_mean_` value (Equations 5 and ??). These are the two outcomes Y_i referenced in the H2 estimand (Equation 7) and in the directional predictions (Equations 8–9). *velocity* corresponds directly to Weinmann’s movement-speed construct (path length over elapsed movement time); *real_ideal_trajectory_diff* captures the same path-deviation construct but as an absolute pixel-scale difference rather than a normalized actual-to-shortest ratio, so it is reported on its native pixel scale and is, by construction, sensitive to screen resolution and pointer gain.

5 Results

5.1 Estimation and Power

Each ATE is the slope on the (0/1) FG indicator in a weighted linear model using the stabilized IPTW weights, with Huber–White HC3 heteroskedasticity-robust standard errors MacKinnon and White (1985). Directional one-sided tests are used where a sign is theoretically pre-specified; otherwise tests are two-sided. We report 95% confidence intervals throughout.

Multiple-comparison adjustment. The two primary cursor outcomes for H2 (trajectory deviation, movement speed) are Holm-adjusted as a confirmatory family. A pre-specified secondary cursor family of 11 features is treated as exploratory and adjusted with Benjamini–Hochberg FDR at $q = .05$. H1 (self-report) and H3 (moderation) involve a single test family each and are reported unadjusted.

Minimum detectable effect (MDE). At $n_1 = 57$, $n_2 = 55$, $\alpha = .05$ (two-sided), and $1 - \beta = .80$, the design’s minimum detectable standardized effect is approximately 0.58 once we account for the modest loss of effective sample size under IPTW (the effective ns fall to 46.2 and 48.8). The design therefore has adequate power for medium-to-large effects but not small effects; we use this MDE ≈ 0.58 benchmark to discipline interpretation of the null findings below.

5.2 H1 Result: Social Desirability (Manipulation Check)

H1 predicts that the FG instruction elevates a composite social-desirability score formed as the mean of BFI-2 Extraversion, Agreeableness, Conscientiousness, and reverse-scored Negative Emotionality. Table 6 reports the IPTW-weighted ATE.

Table 6: H1: IPTW-weighted ATE of the FG instruction on the social-desirability

Outcome	ATE	SE	95% CI	z	p (one-sided)
Social desirability (SD)	+0.23***	0.06	[+0.11, +0.34]	3.92	< .001

*** $p < .001$.

Note. Directional one-sided test (ATE > 0); HC3 robust SE.

H1 is strongly supported. The FG instruction causally elevates the social-desirability composite by +0.227 scale units (SE = 0.058, 95% CI [+0.113, +0.340], $z = 3.92$, one-sided $p < .001$). The estimate is attenuated relative to the unadjusted difference in means ($\hat{\tau}_{\text{unwt}} \approx 0.67$), as expected: IPTW down-weights observations whose covariates predict

the FG arm, and the imbalanced covariates (female, computer skill) are themselves associated with self-report style. The manipulation check is satisfied: the FG instruction induces more socially desirable self-presentation in response *content*, providing the necessary precondition for interpreting H2 estimates as effects of induced faking rather than instructional difference per se.

5.3 H2 Result: Cursor-Dynamic Outcomes

H2 has two pre-specified primary outcomes, each with a directional sign: trajectory deviation (mean realized-minus-ideal path-length difference on the BFI task, in pixels; H2a, predicted > 0) and movement speed (mean velocity; H2b, predicted < 0). Table 7 reports the IPTW-weighted ATEs.

Table 7: H2: IPTW-weighted ATEs on the two primary cursor outcomes.

Outcome	ATE	SE	95% CI	z	p (1-sided)	p_{Holm}
Trajectory deviation (H2a)	-16.16	126.17	$[-263.45, +231.14]$	-0.13	.551	1.000
Movement speed (H2b)	+0.03	0.03	$[-0.02, +0.08]$	+1.05	.854	1.000

Note. Directional one-sided tests and Holm adjustment across the primary family. HC3 robust SE

Primary indicators. Neither primary cursor outcome supports H2. Trajectory deviation is essentially zero with the point estimate in the *opposite* direction to prediction (-16.2 , $p = .551$). Movement speed is positive rather than negative as predicted and does not approach significance ($+0.028$, $p = .854$). Both Holm-adjusted p -values equal 1.00. The 95% CIs straddle zero and rule out medium-to-large effects in the predicted direction. The wide interval on trajectory deviation (± 247 pixels) reflects in part that this feature is an unnormalized pixel-scale difference (Section 3.2), so between-participant variance is inflated by heterogeneous screen resolutions and pointer gains across the in-the-wild sample.

Secondary exploratory family (FDR-adjusted). The dataset additionally exposes a broader family of trial-level mouse dynamics features used in the source paper’s classifier (Kuric et al., 2025, Table 4), spanning initialization latency, total reaction time, area-under-the-curve (AUC) difference, axis-specific direction reversals, and so on. We estimate effects on these as a secondary family with Benjamini–Hochberg FDR adjustment, treating them as exploratory rather than confirmatory in Appendix A. The smallest unadjusted p -value is .054 for the AUC difference, but no feature survives BH adjustment ($p_{\text{BH}, \min} = .593$). The family-wide null reinforces the primary-outcome conclusion; the full per-feature estimates are reported in Table 8.

H2 is therefore *not* supported. Neither primary cursor outcome moves in the predicted direction once pre-treatment imbalance is enforced via IPTW, and the broader exploratory family yields no FDR-surviving effects.

6 Robustness Checks

The logistic propensity-score model used in the main analysis may be mis-specified, so we subject the ATE estimates to three robustness analyses: doubly-robust estimation (covariate-adjusted IPTW and AIPW), three alternative propensity-score weighting schemes (CBPS, entropy balancing, and GBM), and a pairs bootstrap that re-estimates the weights in each resample. All three are reported in full in Appendices C, D, and E) respectively, and all agree with the main analysis: the H1 effect is stable across every check and the H2 cursor outcomes remain null under all of them.

7 Discussion

7.1 Interpreting the three results

Three findings shape our interpretation. First, the faking-good manipulation clearly changed what participants reported: the social-desirability composite rose by about one standard deviation, robust across covariate-adjusted IPTW, AIPW, four propensity-score specifications, and a bootstrap that re-estimated the weights. The FG arm did engage the impression management the design set out to induce. Second, and key, our null sits alongside the source paper’s trial-level classifier ($F1 = 86.9\%$). From our perspective, the high predictive power and the null effect from mouse movement hypothesis testing do not contradict each other because they answer different questions. The source paper asks a *predictive* question: is there some rule, using the cursor features together, that separates honest from faking responses? We ask a narrower, *causal* question: does faking change one specific feature, on average, in the direction theory predicts? A classifier can succeed at the first even when the answer to the second is no—by combining features, exploiting differences in spread rather than mean, or picking up trial-by-trial patterns that vanish once we average each participant down to a single number. Our null neither overturns nor confirms the original result; it narrows how the classifier should be read: whatever it is exploiting, it is probably not the large, stable, directional motor signature that the cognitive-conflict account predicts at the participant level. Third, against this confirmed manipulation, the participant-level cursor signal did not appear: neither pre-specified primary outcome moved in the predicted direction, both point estimates fell

near zero and, where signed, in the *opposite* direction to prediction, and no feature in the broader exploratory set survived multiple-comparison adjustment. As this null pattern is at odds with the directional effects predicted by the response activation model, we examine possible explanations in Section 7.2.

7.2 Why the participant-level cursor signal is absent

We offer three post hoc interpretations, both of which preserve the Response Activation Model (Welsh and Elliott, 2004) and cognitive load theory (Sweller, 1988; Buller and Burgoon, 1996) rather than count against them, and neither of which the present data establishes.

Level mismatch. The Response Activation Model and cognitive load theory describe what happens during a *single conflicted movement*: when two response options are jointly active, the path bends away from the straight line and the hand slows. They do not predict that a participant’s deviation or speed, averaged over every trial, shifts under faking—these are claims at different levels. Most items in a faking session call for little or no fabrication, so only a minority of trials carry real response conflict. When we collapse a participant’s trials into one average, the few conflict-bearing trials are diluted by the many that are not, and whatever signal they carry is washed out. Read this way, our aggregate null and the source paper’s trial-level classifier fit together rather than clash: both point to the same conclusion—if a cursor signal of faking exists, it lives at the level of individual trials, exactly where the theory says it should.

Dose-response in the intention–response mismatch. Weinmann et al. (2022) found that the cursor signature of misreporting grew with the size of the fraud—a dose-response pattern. Faking-good shares the same core ingredient (a gap between honest and reported answer) but lacks the material gain and harmed counter-party that fraud involves (Section 2.2). In dose-response terms, faking-good is a *smaller dose* of that mismatch, and the cognitive-conflict account therefore predicts a *weaker* signature rather than its absence—consistent with point estimates near zero at or below the design’s detection threshold. The null does not contradict the cognitive-conflict account; it supports the very fraud/faking-good distinction motivating our taxonomy and marks the limit of how far fraud-validated metrics carry over to faking-good.

Strength of the interface manipulation. The only difference between faking-good and honest arms was a single line of small-type instruction to present oneself favorably in this study. This was enough to move self-reports—the social-desirability composite

rose by about one standard deviation—but perhaps too weak to change how participants moved the cursor. A one-line prompt can alter the answer someone gives without engaging the moment-to-moment deliberation the cursor records: the manipulation from Kuric et al. (2025) drove self-report but not motor behavior. On this reading, the null reflects a weak manipulation rather than the absence of a cursor signature of faking, and a more demanding interface might recover the mouse-movement and mouse-speed differences that did not surface here.

8 Limitations and Future Work

8.1 Limitations

Our reanalysis is restricted to the aggregated cursor features the source authors exposed; we cannot access the raw trajectories or per-trial conflict structure that a direct test of the trial-level mechanism in Section 7.2 would require. The level-mismatch interpretation is therefore offered, not demonstrated: with only participant-level summaries we can establish that the aggregate signal is absent, but not that a conflict-conditioned trial-level signal is present. Two further limitations—a single Slovak sample with mean age ≈ 24 that leaves cross-population generalization untested, and the underpowered H3 moderation—are detailed in Appendix F.

8.2 Future Work

Finally, returning to our two questions mentioned in ref: the social-desirability score is a robust outcome that can serve as a manipulation check, while the mouse-movement metrics could indicate how well the intervention changes respondents’ behavior in the promotion-focused, persuasive hope-appeal intervention. The three interpretations in Section 7.2 are directly testable, and each suggests a concrete design in the future study in promotion-focused, persuasive hope-appeal intervention.

First, a *trial-level, conflict-conditioned* reanalysis would test the level-mismatch account head-on: with trial-by-trial telemetry, one could model deviation and speed on a pre-specified subset of high-conflict trials—items most diagnostic of the faked trait, or items with the largest honest-to-FG response gap—rather than averaging across the whole questionnaire, asking whether the cursor effect re-emerges on exactly the trials where the mechanism predicts it should.

Second, the intervention should maximize participants’ motivation to fake good, pushing the manipulation further up the *dose-response* curve: even without the material gain or

harmed counter-party of fraud, a stronger pull toward impression management should widen the honest-to-reported gap—and any cursor signature that scales with it.

Third, the *interface manipulation* should be made more salient, so that it shifts not only the social-desirability score but also the cursor behavior it is meant to drive.

Connecting these results and interpretations back to the two questions posed in Section 1: the social-desirability score is a robust outcome that can serve as a manipulation check, while the mouse-movement metrics could indicate how well the intervention changes respondents’ behavior. In designing the planned promotion-focused, persuasive hope-appeal intervention, we should attend to *dose effect* and *interface manipulation* in the experimental setup, and to *level mismatch* when testing the hypotheses.

A H2 Secondary Cursor-Feature Family

This appendix reports the full per-feature estimates for the 11-feature secondary cursor family referenced in Section 5.3. These features were pre-specified as an exploratory family and adjusted with Benjamini–Hochberg FDR at $q = .05$; they are treated as exploratory rather than confirmatory. As summarized in the main text, the smallest unadjusted p -value is .054 (AUC discrepancy) and no feature survives BH adjustment ($p_{\text{BH}, \min} = .593$), reinforcing the primary-outcome null.

Table 8: H2 secondary (exploratory) cursor-feature family

Feature	ATE	SE	z	p	p_{BH}
AUC discrepancy	$+1.23 \times 10^4$	6.39×10^3	+1.93	.054	.593
Initialization time	+124	133	+0.93	.350	.793
Max deviation	+16.8	18.3	+0.92	.357	.793
y-axis reversals	+0.369	0.423	+0.87	.383	.793
x-axis velocity	+0.0308	0.0243	+1.27	.204	.793
x-axis acceleration	$+1.24 \times 10^{-5}$	1.58×10^{-5}	+0.78	.432	.793
Reaction time	−132	307	−0.43	.667	.815
y-axis velocity	+0.00507	0.00982	+0.52	.606	.815
Overall acceleration	$+9.22 \times 10^{-6}$	1.85×10^{-5}	+0.50	.619	.815
x-axis reversals	+0.04468	0.369	+0.13	.899	.902
y-axis acceleration	-1.01×10^{-6}	8.23×10^{-6}	−0.12	.902	.902

Note. Two-sided IPTW-weighted ATEs with Benjamini–Hochberg FDR adjustment; treated as exploratory, not confirmatory.

B H3 Exploratory Moderation Analysis

This appendix develops the pre-specified but exploratory employment-adversity moderation hypothesis (H3), introduced in Section 3.3: its theory and estimand (Appendix B.1),

moderator construction, and results. H3 is underpowered at $n = 112$ and is reported here for completeness and for planning future studies.

B.1 Hypothesis and Estimand

The faking-good manipulation does not act on a uniform population. Levashina and Campion (2007) characterize applicant faking as strategic behavior whose intensity scales with perceived stakes and with accumulated experience navigating selection contexts: faking is likeliest when applicants perceive high stakes *and* hold concrete referents from comparable past situations, while those without such referents engage more abstractly. Bolino et al.’s (2008) review of impression management converges on the same stakes-by-history interaction. Applied here, the dream-job framing supplies the high-stakes cue, and participants who have personally faced employment adversity—long-term unemployment while job-seeking, or involuntary job loss—hold concrete autobiographical referents that should raise both the perceived stakes of the simulated application and the strategic deployment of impression-management tactics. We make no claim that employment history is a stable trait or exerts a main effect on social desirability; the proxy is deliberately minimal and the hypothesis concerns moderation of the FG effect only.

H3 estimand and hypothesis. The moderator $M_i \in \{0, 1\}$ equals 1 if participant i reports long-term unemployment while job-seeking or involuntary job loss, and is measured pre-treatment. For each outcome Y_i (the SD_i , deviation, and speed of H1–H2), the conditional average treatment effect at moderator level m is

$$\text{CATE}_Y(m) = E[Y_i \mid D_i = \text{FG}, M_i = m] - E[Y_i \mid D_i = \text{H}, M_i = m]. \quad (13)$$

Because D_i is randomized within each stratum of M_i , $\text{CATE}_Y(m)$ is causally identified at each level (VanderWeele, 2009), though M_i itself is observational. The moderation contrast

$$\Delta_Y = \text{CATE}_Y(1) - \text{CATE}_Y(0) \quad (14)$$

is estimated via a treatment $\times$ moderator interaction in a participant-level regression of Y_i on D_i , M_i , and their product. We predict

$$\text{H3: } \Delta_Y \neq 0, \quad (15)$$

directionally $\Delta_{SD} > 0$, $\Delta_{\text{Deviation}} > 0$, and $\Delta_{\text{Speed}} < 0$: participants with prior employment adversity are expected to show a larger shift toward socially desirable responding and a stronger cursor signature of faking under FG. We designate H3 exploratory: at $n = 112$ the interaction test is substantially underpowered (Gelman, 2015; Simmons

et al., 2011), so we report cell sizes transparently and do not read underpowered nulls as evidence against moderation.

B.2 Moderator Construction

The binary moderator $M_i \in \{0, 1\}$ in Equations 13–14 captures prior employment adversity and is constructed from three behaviorally-anchored items capturing employment adversity from the source paper’s validation questionnaire: (i) ever long-term unemployed (more than six months while actively seeking work), (ii) ever involuntarily terminated, and (iii) currently unemployed. Items index the referent-activation mechanism articulated above (Appendix B.1)—each supplies concrete autobiographical referents from consequential past selection contexts (Levashina and Campion, 2007; Bolino et al., 2008)—so we had no theoretical basis for treating them as distinct moderators a priori. To maximize the $M = 1$ subsample given the small overall sample, we pre-specified that the three items would be combined via logical OR into a single binary moderator M , coded 1 if any item was positive and 0 otherwise.

Table 9 reports the realized counts. To maximize the $M = 1$ subsample for subgroup inference, we combined three behaviorally-anchored items capturing employment adversity via logical disjunction (OR): (i) ever long-term unemployed while actively seeking work, (ii) ever involuntarily terminated, and (iii) currently unemployed at the time of the study. Each item individually yielded too few positive responses for standalone subgroup analysis—12, 4, and 2 positive responses, respectively—and the contribution of “ever fired” and “currently unemployed” to the Faking-Good $M = 1$ count was negligible because employment adversity is structurally rare in this arm. The combined indicator M yields 17 positive responses (11 in H, 6 in FG), producing 2×2 cell counts ($H/\text{no}, H/\text{yes}, FG/\text{no}, FG/\text{yes}$) = (46, 11, 49, 6). Despite the OR aggregation, the smallest cell still contains only 6 participants—all in the Faking-Good arm—which is well below thresholds for meaningful subgroup inference. We therefore mark H3 as exploratory and refrain from substantive interpretation of any moderation estimates Gelman (2015); Simmons et al. (2011).

Table 9: Positive responses for the three H3 moderator items and the combined indicator M , by treatment arm.

Item	Honest	Faking-Good	Total
Ever long-term unemployed	9	3	12
Ever fired	1	3	4
Currently unemployed	2	0	2
Combined moderator M	11	6	17

B.3 Moderation Results

H3 asks whether the FG treatment effect on each of the three primary outcomes differs by prior employment adversity. The combined moderator $M \in \{0, 1\}$ takes value 1 if the participant has ever been long-term unemployed, ever been fired, or is currently unemployed; it is positive for 17 participants (H: 11, FG: 6). We estimate

$$Y_i = \alpha + \tau \text{FG}_i + \beta_M M_i + \gamma (\text{FG}_i \times M_i) + \varepsilon_i, \quad (16)$$

in the IPTW-weighted sample with HC3 robust SE. Because FG is randomized within each level of M_i VanderWeele (2009), γ identifies a causal moderation of the FG treatment effect. All tests are two-sided.

Table 10: H3 (exploratory): IPTW-weighted moderation by employment adversity.

Outcome	Δ_Y	SE	z	p
Social desirability	+0.08	0.14	+0.58	.559
Trajectory deviation	−685.92	507.79	−1.35	.177
Movement speed	−0.16	0.10	−1.56	.118

Note. Δ_Y is the $\text{FG} \times M$ interaction coefficient, i.e. $\text{ATE}(M = 1) - \text{ATE}(M = 0)$. HC3 robust SE.

No interaction reaches significance. The two cursor interactions are nominally larger in magnitude than the SD interaction, but the moderator-by-arm cells are very small ($\text{FG}/M = 1$ contains only 6 participants), so the wide CIs preclude substantive interpretation. We treat H3 as null-informative for future power calculations rather than as a theoretical contribution.

C Doubly-Robust Alternatives

The IPTW estimator from Section 4.6 is consistent only when the logistic propensity-score model is correctly specified. Because that assumption is not directly verifiable, we re-estimate the ATE with two doubly-robust (DR) alternatives: covariate-adjusted IPTW and Augmented Inverse Propensity Weighting (AIPW).

The intuition for why a DR estimator improves on plain IPTW is the following. IPTW relies on a single model—the propensity score—and is biased whenever that model is wrong. A DR estimator brings in a second, independent model on the outcome side, and combines the two in such a way that the estimator remains consistent as long as *at least one of the two models is correct*. The analyst therefore has two chances to be right instead of one. Covariate-adjusted IPTW is the simplest realisation of this idea: it keeps the IPTW weights *and* enters the six covariates as additional regression controls in the

weighted outcome model, so a mis-specified propensity score can still be rescued by the outcome regression.

The doubly-robust property of covariate-adjusted IPTW, however, hinges on the outcome model being *linear* in the covariates Robins et al. (1992); Wooldridge (2007). If the true outcome–covariate relationship is non-linear, this linearity-based DR guarantee weakens and the estimator can behave more like plain IPTW than like a fully doubly-robust estimator. AIPW is the canonical DR construction and does not depend on outcome linearity in the same way, so it serves as a more theoretically robust check. Reading the three estimates side by side then yields a clean interpretation: if IPTW, covariate-adjusted IPTW, and AIPW all agree, the IPTW result from Section 4.6 is trustworthy and no correction is needed. If AIPW disagrees with the other two, this signals that the IPTW weighting used in the main analysis is unreliable for this dataset, and the AIPW estimate should be adopted as the more credible answer. Table 11 places the three estimators—IPTW, covariate-adjusted IPTW, and AIPW—side by side for each of the three primary outcomes.

Table 11: Doubly-robust robustness check

Outcome	Estimator	ATE	SE	95% CI	z	p
Social desirability (H1)	IPTW	+0.227	0.058	[+0.113, +0.340]	3.92	< .001
	cov-adj IPTW	+0.235	0.052	[+0.133, +0.337]	4.52	< .001
	AIPW	+0.239	0.047	[+0.148, +0.331]	5.13	< .001
Trajectory deviation (H2a)	IPTW	−16.156	126.173	[−263.451, +231.138]	−0.13	.551
	cov-adj IPTW	−28.564	125.924	[−275.371, +218.242]	−0.23	.590
	AIPW	−59.697	118.967	[−292.867, +173.474]	−0.50	.692
Movement speed (H2b)	IPTW	+0.028	0.027	[−0.024, +0.081]	1.05	.854
	cov-adj IPTW	+0.032	0.026	[−0.018, +0.082]	1.25	.895
	AIPW	+0.031	0.024	[−0.015, +0.078]	1.31	.905

Note. IPTW (propensity-score weighting only), cov-adj IPTW (covariate-adjusted IPTW), and AIPW (augmented, influence-function SE) for the three primary outcomes. Directional tests: H1 and H2a one-sided > 0 , H2b one-sided < 0 .

The three estimators converge on the same answer for every primary outcome (Table 11). For H1, IPTW (+0.227), covariate-adjusted IPTW (+0.235), and AIPW (+0.239) agree to within 0.012, with all three 95% CIs lying entirely above zero. Importantly, AIPW is slightly *larger* than IPTW rather than smaller, so the doubly-robust correction does not pull the estimate back toward the unweighted ~ 0.67 ; both DR and AIPW therefore corroborate the IPTW shrinkage as a genuine confounding correction driven by the sex and computer-skill imbalances identified in Section 4.5, rather than as over-correction by a mis-specified propensity-score model. For H2, the three estimators agree just as cleanly on a null: trajectory deviation (H2a) and movement speed (H2b) show no detectable effect under any estimator, every 95% CI straddles zero, and the null is therefore robust to the choice between weighting-only, covariate-adjusted IPTW, and fully augmented

AIPW.

D Alternative Propensity-Score Specifications

The AIPW result in Appendix C protects against propensity-score mis-specification *only when the outcome model is correctly specified*. If both nuisance models share the same blind spot—most plausibly, both being linear in the covariates while the underlying relationships are not—the doubly-robust guarantee no longer applies. We therefore re-estimate the ATE under three alternative weighting schemes whose propensity-score specifications do *not* rely on the logistic-additive form used in the main analysis.

The main IPTW model implicitly assumes that (i) each covariate’s effect on the treatment probability is linear on the logit scale, (ii) covariate effects are additive with no interactions, and (iii) the continuous covariates require no non-linear transformation. These are strong assumptions for the two continuous covariates in our design. Computer skill, for example, plausibly relates to cursor behavior non-linearly: novices are slow, experts are fast, and the gain from one additional scale point may grow closer to exponentially than linearly across the 1–5 range. When such a non-linearity exists in the covariate–outcome relationship but the propensity score is fitted as linear-additive, balancing covariate *means*—as the Love plot in Section 4.5 confirms—is not sufficient; higher moments remain imbalanced and continue to contaminate the ATE.

Three alternative weighting schemes relax the logistic-additive form in different ways:

- **Covariate Balancing Propensity Score (CBPS)** Imai and Ratkovic (2014)—still a parametric logistic specification, but estimated by directly targeting covariate balance through a generalized method-of-moments objective rather than by maximum likelihood. CBPS therefore avoids the finite-sample bias that likelihood-based estimation can suffer when the propensity-score model is mildly mis-specified.
- **Entropy Balancing** Hainmueller (2012)—abandons the propensity-score model entirely and instead computes weights that force the treatment and control groups to match exactly on pre-specified covariate moments. Because no functional form is imposed on the assignment mechanism, entropy balancing cannot be wrong about the propensity-score form.
- **Generalized Boosted Models (GBM)** McCaffrey et al. (2004)—a fully non-parametric machine-learning estimator that fits the propensity score as an ensemble of regression trees. GBM automatically captures non-linearities and interactions among the covariates without requiring the analyst to pre-specify them.

The four weighting schemes (logistic GLM, CBPS, entropy balancing, GBM) share no common functional-form blind spot. If all four return similar ATE estimates, the IPTW conclusion is robust to the propensity-score specification; if a more flexible scheme (in particular GBM) diverges from the logistic-additive estimate, the linearity assumption of the main analysis is likely violated and the more flexible estimate should be preferred. Table 12 shows the ATE with 95% CI from the four propensity-score specifications—logistic GLM, CBPS, entropy balancing and GBM.

Table 12: Alternative propensity-score specifications

PS method	H1: SD	H2a: Deviation	H2b: Speed
Logistic GLM (main)	+0.227 [+0.113, +0.340]	-16.156 [-263.451, +231.138]	+0.028 [-0.024, +0.081]
CBPS	+0.237 [+0.121, +0.352]	-24.022 [-272.872, +224.828]	+0.034 [-0.020, +0.088]
Entropy balancing	+0.247 [+0.129, +0.364]	-58.120 [-327.461, +211.221]	+0.032 [-0.025, +0.089]
GBM	+0.218 [+0.110, +0.326]	-36.001 [-262.014, +190.013]	+0.023 [-0.027, +0.073]

Note. H1: SD, Socially Desirable outcome; H2a Deviation, Mouse movement deviation; H2b: Speed, mouse movement speed.

The four weighting schemes produce a consistent picture across all three primary outcomes (Table 12): H1 point estimates cluster in [+0.218, +0.247] with every 95% CI above zero, while H2a and H2b estimates all carry the wrong sign but every CI straddles zero by a wide margin. The decisive comparison is GBM—which imposes no functional form—versus the logistic-GLM main analysis: GBM returns +0.218, less than 0.01 below the main estimate and inside one standard error, so if linearity had been biasing the IPTW result, the bias would have surfaced here. CBPS and entropy balancing actually pull H1 slightly *upward*, suggesting the main estimate is, if anything, mildly conservative. The functional-form assumption of the main analysis is therefore not driving either the H1 significance or the H2 null.

E Bootstrap Inference

A separate concern is that the HC3 standard errors reported in Section 4.6 treat the IPTW weights as fixed, known quantities, when in fact they are estimated from the same data using a logistic propensity-score model. This issue is particularly salient for H1: the point estimate moved from the unweighted difference of approximately 0.67 to the IPTW estimate of approximately 0.23, meaning that nearly two thirds of the apparent raw effect was absorbed by the estimated weights. The propensity-score model is therefore doing substantial work in the H1 analysis, and any uncertainty in the weights themselves should propagate into the standard error of the ATE. HC3 does not propagate it: the weights enter the weighted regression as if they were observed covariates rather than estimated

quantities, which can understate the true uncertainty of the ATE and yield confidence intervals that are too narrow for the procedure as a whole Austin (2022).

We address this with a non-parametric pairs bootstrap in which the propensity score is re-estimated within each resample Efron (1979); Austin (2022). The procedure mirrors the analytic pipeline end to end: in each of $B = 2,000$ replications we (i) draw $n = 112$ observations with replacement from the original sample, (ii) refit the logistic propensity-score model on the resampled data and recompute stabilised weights, (iii) fit the weighted outcome regression on the resample, and (iv) record the ATE coefficient on the Faking-Good indicator. The standard deviation of the resulting 2,000 ATE estimates is the bootstrap standard error, and the 2.5th and 97.5th percentiles form the bootstrap percentile 95% confidence interval. Because the propensity score is re-estimated in every replication, the resulting interval captures both sources of sampling variability that HC3 ignores by construction: variability in the outcome regression and variability in the estimated weights themselves. If the bootstrap and HC3 intervals agree, the analytic inference is validated; if the bootstrap interval is materially wider, the bootstrap interval should be adopted as the primary inference. Table 13 demonstrates pairs bootstrap with propensity-score re-estimation in each resample.

Table 13: Bootstrap inference

Outcome	ATE	HC3 95% CI	Boot SE	Boot 95% CI	B
Social desirability (H1)	+0.227	[+0.113, +0.340]	0.054	[+0.121, +0.328]	2,000
Trajectory deviation (H2a)	-16.156	[-263.451, +231.138]	123.457	[-268.481, +200.574]	2,000
Movement speed (H2b)	+0.028	[-0.024, +0.081]	0.024	[-0.022, +0.074]	2,000

The bootstrap matches the HC3 inference for every primary outcome (Table 13): H1 stays well above zero under both methods ([+0.121, +0.328] vs. [+0.113, +0.340]), and H2a and H2b both straddle zero by wide margins. Notably, the bootstrap intervals are about 8–9% *narrower* than HC3 in every row, not wider—the opposite of what one would expect if the estimated weights were a meaningful source of extra variance. The weights are stable enough across resamples that the additional variance from re-estimating them is negligible, and HC3—which treats the weights as fixed—is an adequate approximation here. The strong H1 result is therefore not a small-sample artifact: it survives even when the inference fully propagates the uncertainty in the estimated weights.

F Additional Limitations

This appendix expands the limitations summarized in Section 8.1.

Single population. The analysis draws on one secondary dataset from one population—a predominantly young Slovak sample (mean age ≈ 24)—so external validity across populations and instruments is untested here. This limitation is sharpened by the dose-response interpretation in Section 7.2: because that account makes the cursor signature contingent on the perceived stakes of the faking task, and the salience of a “dream-job” framing plausibly varies across labor markets, age cohorts, and cultural contexts, generalization beyond the present sample is explicitly an open question rather than a safe default.

Underpowered moderation (H3). The H3 analysis of how employment adversity moderates the faking-good effect (Appendix B) is badly underpowered. The smallest cell formed by crossing treatment with the moderator contains only six participants, and the moderator itself was measured rather than randomized. We therefore attach no substantive weight to its non-significant interactions; H3 is informative only as a benchmark for planning the sample size of future studies (Gelman, 2015; Simmons et al., 2011).

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